The Alaoglu–Erdős Conjecture

Mesoscopic Factorial Cocycles,

Prime-Window Obstructions, and a Linear-Width Escape Target

A nonduplicative supplemental progress report

to the unified ProveIt report supplied on August 22, 2026

Bottom line. I did not obtain a complete proof of

$$2^x, 3^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z}.$$

I did obtain a new, exact attack that is not developed in the supplied report, together with an essentially complete diagnosis of its fixed-width and subcritical-width forms. For a hypothetical counterexample put $M = 2^x$, $A = 3^x$, and $N_n = A^n$, $K_n = M^n$. The universal identity

$$K_n = N_n^{\log 2/\log 3}, \qquad \frac{1}{2} < \frac{\log 2}{\log 3} < \frac{2}{3}$$

places the falling-factorial probability

$$R_n = \frac{(A^n)_{M^n}}{(A^n)^{M^n}}$$

in a uniquely favorable "birthday" regime: pair collisions diverge as $(M^2/A)^n = (4/3)^{xn}$, while triple collisions vanish as $(M^3/A^2)^n = (8/9)^{xn}$. A finite shift polynomial can cancel the sole growing collision mode exactly.

For every integer polynomial P, the signed factorial cocycle

$$\mathcal{F}_{P,n} = \prod_j \left(\frac{A^{n+j}!}{(A^{n+j} - M^{n+j})!} \right)^{[T^j]P}$$

tends to 1 precisely when

$$P(M) = P'(M) = P(M^2/A) = 0.$$

The primitive minimal annihilator is

$$P_*(T) = (T - M)^2 \left(\frac{A}{g} T - \frac{M^2}{g} \right), \qquad g = \gcd(A, M^2),$$

and its cocycle satisfies the sharp asymptotic

$$\log \mathcal{F}_{P_*,n} \sim \frac{M^2(A - M)}{6gA} \left(M - \frac{M^3}{A^2} \right)^2 \left(\frac{M^3}{A^2} \right)^n > 0.$$

Thus it is a positive rational number converging exponentially to 1.

The arithmetic side is hostile in an equally exact way. Using the inherited exclusion $x \geq 12$, Huxley's prime number theorem in short intervals, and Brun–Titchmarsh covering, I prove that every such cocycle whose shifts extend less than

$$\bigl(\log_2 3 - 1 - \varepsilon\bigr) n = \bigl(0.5849625007\ldots - \varepsilon\bigr) n$$

beyond its highest negative coefficient has a reduced denominator with logarithm $\gg M^n$. In particular, every fixed-shift factorial or binomial cocycle fails by an exponentially large prime-window denominator. The first possible escape is therefore not "improve a constant" but a sharply specified linear-width prime-transport problem. I give its exact valuation inequalities, two phase thresholds, a finite optimization formulation, and a Lean-oriented decomposition.

Prepared for Vladimir Reshetnikov

August 22, 2026

Abstract

Let $x > 0$ and suppose that $M = 2^x$ and $A = 3^x$ are integers. The supplied unified report already develops the exact conditional solution monoid, local and global transcendence barriers, generalized Vandermonde determinants, p-adic and Kummer diagnostics, compact-orbit models, entire interpolation, Stieltjes moments, carry languages, and numerous quantified no-go theorems. This supplement deliberately avoids repeating those constructions.

The new object is the mesoscopic falling factorial $F_n = (A^n)_{M^n}$ and its normalized no-collision probability $R_n = F_n/(A^n)^{M^n}$. Since $\alpha = \log M/\log A = \log 2/\log 3 \in (1/2, 2/3)$, one has $M^{2n}/A^n \to \infty$ but $M^{3n}/A^{2n} \to 0$. An explicit logarithmic expansion gives

$$\log F_n = nM^n \log A - \frac{1}{2}\left(\frac{M^2}{A}\right)^n + \frac{1}{2}\left(\frac{M}{A}\right)^n - \frac{1}{6}\left(\frac{M^3}{A^2}\right)^n + O\left(\left(\frac{M^4}{A^3}\right)^n\right).$$

For a polynomial $P(T) = \sum c_j T^j$, the cocycle $\mathcal{F}_{P,n} = \prod_j F_{n+j}^{c_j}$ therefore tends to one if and only if P has a double zero at M and a zero at M^2/A. This yields a canonical minimal near-unit and a complete fixed-shift classification.

The denominator is analyzed prime by prime through exact factorial-valuation floor sums. Every annihilator has at least three coefficient sign variations. If k is its highest negative shift, primes in the interval $(A^{n+k} - M^{n+k}, A^{n+k}]$ enter the denominator unless a higher positive factorial interval captures a multiple. A Brun–Titchmarsh covering estimate shows that, for the least hypothetical counterexample (hence $x \geq 12$), higher shifts of total span below $(\log_2 3 - 1 - \varepsilon)(n + k)$ can capture less than a fixed fraction of the short-interval prime mass. Huxley's theorem supplies total logarithmic prime mass asymptotic to M^{n+k}, so the reduced denominator has exponential logarithmic size. The same obstruction applies to binomial cocycles; their near-unit classification adds a double zero at 1.

The resulting research target is concrete: construct a linearly widening signed factorial ratio, satisfying three archimedean annihilation constraints and all prime-power valuation inequalities, with shift span at least $(\log_2 3 - 1)n$ but sufficiently small coefficient height. This is expressed as an exact integer linear feasibility problem and separated into finite, analytic, and formalizable layers.

Contents

1 Scope, trust boundary, and nonduplication

1.1 What is inherited

The attached unified report is treated as the baseline, not as material to be rewritten[2]. In particular, this supplement imports the following established facts and conventions:

1. a putative nonintegral solution is positive and transcendental;
2. after passing to the least nonintegral solution one has an exact primitive output pair $(M, A) = (2^x, 3^x)$ and an exact rank-two conditional solution monoid;
3. the verified finite exclusion $2^x < 4096$ implies that a least nonintegral solution satisfies $x \geq 12$;
4. the two-base problem is an exact 2×2 four-exponentials boundary, so no assertion below silently invokes four exponentials or Schanuel;
5. determinant, p-adic, Kummer, compact-orbit, rational-function, entire-interpolation, moment, and carry-language routes already have detailed audits in the baseline report.

Only item 3 enters the strongest new denominator theorem. The elementary factorial classification itself holds for every positive integral pair $1 < M < A$ and does not use the real-power origin of that pair. The historical conjecture itself originates in the work of Alaoglu and Erdős on highly composite and related numbers[1].

1.2 What is new here

The baseline report does discuss factorial divisors of interpolation determinants and Smith profiles. Those are not the construction below. Here factorials arise from the occupancy product

$$(N)_K = N(N-1)\cdots(N-K+1),$$

with the special geometric scaling $(N, K) = (A^n, M^n)$. The central new features are:

1. the exact mesoscopic window $K = N^\alpha$ with $1/2 < \alpha < 2/3$;
2. a shift-polynomial calculus that classifies every fixed factorial near-unit;
3. the primitive minimal annihilator P_* and its exact leading defect;
4. a prime-window theorem forcing denominator mass for all fixed and subcritical linearly widening shifts;
5. the universal escape width $\log_2 3 - 1$ and a finite valuation-feasibility target.

Status. **[proved in this report]** for all statements proved in Sections 2–6, except for the explicitly cited short-interval prime theorem and Brun–Titchmarsh inequality. Numerical tables are **[exact computation]**.

1.3 Competitive-status note

The public material located during this run did not contain an inspectable proof text or Lean repository for the Alaoglu–Erdős conjecture. The reported competing route is therefore treated as unavailable information: no mathematical step below is inferred from it, and every claim in this report is auditable independently of the rumor.

2 The mesoscopic exponent forced by the bases two and three

Assume throughout that

$$M = 2^x \in \mathbb{N}, \qquad A = 3^x \in \mathbb{N}, \qquad x > 0. \tag{1}$$

Then $1 < M < A$ and

$$\alpha := \frac{\log M}{\log A} = \frac{\log 2}{\log 3}, \qquad \vartheta := \alpha^{-1} = \log_2 3. \tag{2}$$

The two elementary inequalities $4 > 3$ and $8 < 9$ give

$$\frac{1}{2} < \alpha < \frac{2}{3}, \qquad 1 < \vartheta < 2. \tag{3}$$

Numerically,

$$\begin{aligned} \alpha &= 0.6309297535714574\ldots, \\ 1-\alpha &= 0.3690702464285426\ldots, \\ \vartheta - 1 &= 0.5849625007211563\ldots, \\ 2-\vartheta &= 0.4150374992788437\ldots. \end{aligned} \tag{4}$$

Put

$$N_n = A^n, \qquad K_n = M^n = N_n^{\alpha}, \qquad q := \frac{M}{A} = \left(\frac{2}{3}\right)^x. \tag{5}$$

Three ratios control everything that follows:

$$\lambda := \frac{M^2}{A} = \left(\frac{4}{3}\right)^x > 1, \qquad \eta := \frac{M^3}{A^2} = \left(\frac{8}{9}\right)^x < 1, \qquad \tau := \frac{M^4}{A^3} = \left(\frac{16}{27}\right)^x < q < \eta. \tag{6}$$

Thus

$$\frac{K_n^2}{N_n} = \lambda^n \longrightarrow \infty, \qquad \frac{K_n^3}{N_n^2} = \eta^n \longrightarrow 0. \tag{7}$$

Observation 2.1 (The one-divergent-cumulant regime). For occupancy of K_n labeled balls in N_n boxes, the pair-collision scale diverges while the triple-collision scale vanishes. This happens because the fixed exponent $\alpha = \log 2/\log 3$ lies strictly between $1/2$ and $2/3$. No estimate involving the unknown size of x is needed.

The inherited finite exclusion becomes useful through q. For a least nonintegral solution, $x \geq 12$, and therefore

$$q \leq q_{12} := \left(\frac{2}{3}\right)^{12} = 0.007707346629258934\ldots. \tag{8}$$

This small ratio will leave almost all short-interval primes uncaptured by higher factorial windows.

3 The birthday product and its exact asymptotic expansion

3.1 Falling factorial and no-collision probability

Define

$$F_n := (A^n)_{\underline{M^n}} = \frac{A^n!}{(A^n - M^n)!} \in \mathbb{N}, \qquad R_n := \frac{F_n}{(A^n)^{M^n}}. \tag{9}$$

Equivalently,

$$R_n = \prod_{j=0}^{M^n-1} \left(1 - \frac{j}{A^n}\right). \tag{10}$$

It is the probability that M^n independent uniform samples from an A^n-element set are all distinct. The probability language is only mnemonic; all arguments use the exact finite product.

3.2 Two-term expansion with a geometric remainder

Lemma 3.1 (Uniform logarithmic remainder)**.** *For $0 \le u < 1$,*

$$0 \le -\log(1-u) - u - \frac{u^2}{2} \le \frac{u^3}{3(1-u)}. \tag{11}$$

Proof. Expand $-\log(1-u) = \sum_{r\ge1} u^r/r$. The remaining terms are nonnegative, and for $r \ge 3$ one has $1/r \le 1/3$, giving the geometric upper bound. □

Proposition 3.2 (Exact mesoscopic expansion)**.** *As $n \to \infty$,*

$$\log F_n = nM^n \log A - \frac{1}{2}\lambda^n + \frac{1}{2}q^n - \frac{1}{6}\eta^n + O(\tau^n). \tag{12}$$

More explicitly,

$$\log F_n = nM^n \log A - \frac{M^n(M^n-1)}{2A^n} - \frac{M^n(M^n-1)(2M^n-1)}{12A^{2n}} + E_n, \tag{13}$$

where, once $q^n \le 1/2$,

$$|E_n| \le \frac{M^{4n}}{12A^{3n}(1-q^n)}. \tag{14}$$

Proof. Take logarithms in (10) and apply Lemma 3.1 with $u = j/A^n$. The first two power sums are

$$\sum_{j=0}^{K-1} j = \frac{K(K-1)}{2}, \qquad \sum_{j=0}^{K-1} j^2 = \frac{K(K-1)(2K-1)}{6}.$$

For the remainder, $j/A^n \le q^n$ and $\sum_{j<K} j^3 = [K(K-1)/2]^2 < K^4/4$, which gives (14). Expanding the two displayed polynomials in $K = M^n$ gives

$$\begin{aligned} -\frac{K(K-1)}{2N} &= -\frac{1}{2}\lambda^n + \frac{1}{2}q^n, \\ -\frac{K(K-1)(2K-1)}{12N^2} &= -\frac{1}{6}\eta^n + \frac{1}{4}q^{2n} - \frac{1}{12}\left(\frac{M}{A^2}\right)^n. \end{aligned}$$

Both omitted bases are smaller than $\tau = M^4/A^3$, while (14) is $O(\tau^n)$. □

Remark 3.3. The expansion is unusually clean: λ^n is the only growing correction to the bulk term $nM^n \log A$. Every collision mode of order at least three decays. This is the exact asymmetry that the shift operator will exploit.

4 Shift-polynomial calculus and complete fixed-shift classification

4.1 Polynomial shifts

Let E denote the forward shift on sequences, $(\mathsf{E}u)_n = u_{n+1}$. For

$$P(T) = \sum_{j=0}^{d} c_j T^j \in \mathbb{Z}[T] \tag{15}$$

define

$$(P(\mathsf{E})u)_n := \sum_{j=0}^{d} c_j u_{n+j} \tag{16}$$

and the positive rational factorial cocycle

$$\mathcal{F}_{P,n} := \prod_{j=0}^{d} F_{n+j}^{c_j}. \tag{17}$$

Negative coefficients mean reciprocals. The logarithm linearizes the construction:

$$\log \mathcal{F}_{P,n} = P(\mathsf{E})(\log F)_n. \tag{18}$$

Two elementary identities drive the classification.

Lemma 4.1 (Shift evaluation on exponential-polynomial modes)**.** *For every $\rho > 0$,*

$$P(\mathsf{E})(\rho^n) = P(\rho)\rho^n, \tag{19}$$
$$P(\mathsf{E})(n\rho^n) = \rho^n(nP(\rho) + \rho P'(\rho)). \tag{20}$$

Proof. The first identity is immediate. For the second,

$$\sum_j c_j(n+j)\rho^{n+j} = \rho^n \left(n \sum_j c_j \rho^j + \sum_j j c_j \rho^j \right) = \rho^n(nP(\rho) + \rho P'(\rho)).$$

□

Theorem 4.2 (Master asymptotic for a fixed shift polynomial)**.** *For fixed $P \in \mathbb{Z}[T]$,*

$$\begin{aligned} \log \mathcal{F}_{P,n} = \log A\, M^n(nP(M) + MP'(M)) - \frac{1}{2}P(\lambda)\lambda^n + \frac{1}{2}P(q)q^n \\ - \frac{1}{6}P(\eta)\eta^n + O_P(\tau^n). \end{aligned} \tag{21}$$

Proof. Apply Lemma 4.1 term by term to Proposition 3.2. Since P is fixed, shifting a geometric $O(\tau^n)$ remainder through the finitely many indices $0, \ldots, d$ preserves its order. □

4.2 Every fixed factorial near-unit

Theorem 4.3 (Near-unit classification)**.** *Let $P \in \mathbb{Z}[T]$. Then*

$$\mathcal{F}_{P,n} \longrightarrow 1 \qquad (n \to \infty) \tag{22}$$

if and only if

$$P(M) = 0, \qquad P'(M) = 0, \qquad P\left(\frac{M^2}{A}\right) = 0. \tag{23}$$

Equivalently, M is a double root of P and M^2/A is a root.

Proof. If the three conditions hold, the first two potentially growing lines of (21) vanish and all remaining terms tend to zero; hence $\log \mathcal{F}_{P,n} \to 0$.

Conversely, suppose $\log \mathcal{F}_{P,n} \to 0$. Since $M > \lambda > 1$, a nonzero $nP(M)M^n$ term cannot be canceled by any later term, so $P(M) = 0$. The remaining bulk term is $\log A\, M P'(M)M^n$; it forces $P'(M) = 0$. The only remaining growing mode is $-P(\lambda)\lambda^n/2$, so $P(\lambda) = 0$. □

Observation 4.4. The theorem is a classification, not merely a construction. Every finite-shift factorial ratio converging to one must use exactly the same three annihilations. A different-looking fixed product cannot evade them.

4.3 The primitive minimal annihilator

Let

$$g := \gcd(A, M^2), \qquad a_0 := \frac{A}{g}, \qquad b_0 := \frac{M^2}{g}. \tag{24}$$

Then $\gcd(a_0, b_0) = 1$ and $\lambda = b_0/a_0$. Define

$$P_*(T) := (T - M)^2(a_0 T - b_0). \tag{25}$$

In coefficient order from low to high,

$$P_*(T) = -\frac{M^4}{g} + \frac{M^2(A + 2M)}{g}T - \frac{M^2 + 2AM}{g}T^2 + \frac{A}{g}T^3. \tag{26}$$

Proposition 4.5 (Minimality and divisibility)**.** *The polynomial P_* is primitive. A polynomial $P \in \mathbb{Z}[T]$ satisfies (23) if and only if*

$$P_*(T) \mid P(T) \quad \textit{in } \mathbb{Z}[T]. \tag{27}$$

In particular, every nonzero fixed factorial near-unit has degree at least three.

Proof. The factors $(T - M)^2$ and $a_0 T - b_0$ are primitive and coprime in $\mathbb{Q}[T]$ because $M \neq M^2/A$. The rational-root theorem and Gauss's lemma give the claimed divisibility. Primitivity follows again from Gauss's lemma. □

Proposition 4.6 (Sharp defect of the minimal cocycle)**.** *Put*

$$C_* := \frac{M^2(A-M)}{6gA}\left(M - \frac{M^3}{A^2}\right)^2 > 0. \tag{28}$$

Then

$$\log \mathcal{F}_{P_*,n} = C_*\eta^n + O(q^n), \qquad \mathcal{F}_{P_*,n} - 1 \sim C_*\eta^n. \tag{29}$$

In particular, $\mathcal{F}_{P_*,n} > 1$ for all sufficiently large n.

Proof. The three growing terms vanish. Since $\eta > q > \tau$, the leading remaining term in (21) is $-P_*(\eta)\eta^n/6$. Now

$$a_0\eta - b_0 = \frac{A}{g}\frac{M^3}{A^2} - \frac{M^2}{g} = -\frac{M^2(A-M)}{gA},$$

so $-P_*(\eta)/6 = C_*$. Finally $e^u - 1 \sim u$ as $u \to 0$. □

4.4 Three sign changes are unavoidable

Proposition 4.7 (Sign-variation obstruction)**.** *Every nonzero polynomial $P \in \mathbb{Z}[T]$ satisfying (23) has at least three sign variations in its sequence of nonzero coefficients. In particular it has both positive and negative coefficients.*

Proof. The polynomial has the three positive roots $M, M, M^2/A$, counted with multiplicity. Descartes' rule of signs gives at least three coefficient sign variations. □

This simple fact is arithmetically decisive. Near-unit cancellation forces reciprocals of some factorial intervals, and those negative intervals expose primes which positive intervals must somehow transport across geometric scales.

5 Exact denominator valuations

5.1 The floor-sum formula

For a positive rational y, let $\mathrm{den}(y)$ be the positive denominator of y in lowest terms. For $t \geq 0$ and a prime p define

$$V_p(t) := v_p(F_t) = v_p(A^t!) - v_p((A^t - M^t)!). \tag{30}$$

Legendre's formula gives

$$V_p(t) = \sum_{r\geq 1}\left(\left\lfloor \frac{A^t}{p^r}\right\rfloor - \left\lfloor \frac{A^t - M^t}{p^r}\right\rfloor\right). \tag{31}$$

The inner difference counts multiples of p^r in the terminal interval

$$I_t := (A^t - M^t, A^t]. \tag{32}$$

Proposition 5.1 (Exact prime-power valuation of a cocycle)**.** *For* $P(T) = \sum_{j=0}^{d} c_j T^j$*,*

$$\begin{aligned} v_p(\mathcal{F}_{P,n}) &= \sum_{j=0}^{d} c_j V_p(n+j) \\ &= \sum_{r\geq 1}\sum_{j=0}^{d} c_j \left(\left\lfloor \frac{A^{n+j}}{p^r} \right\rfloor - \left\lfloor \frac{A^{n+j} - M^{n+j}}{p^r} \right\rfloor \right). \end{aligned} \tag{33}$$

Consequently

$$\log \operatorname{den}(\mathcal{F}_{P,n}) = \sum_p \max\{0, -v_p(\mathcal{F}_{P,n})\} \log p. \tag{34}$$

Proof. The first identity is additivity of p-adic valuation, the second is Legendre's formula, and the last is the prime factorization of a reduced denominator. □

Remark 5.2 (Finite feasibility)*.* For fixed M, A, n, d, all sums in (33) are finite and exact. Thus the conditions

$$\sum_j c_j V_p(n+j) \geq 0 \qquad (p \leq A^{n+d}) \tag{35}$$

are a finite system of homogeneous integer linear inequalities in the coefficient vector $(c_0, \ldots, c_d)$. The archimedean annihilations are three homogeneous linear equations after clearing the denominator of M^2/A. This will become the finite search target in Section 9.

5.2 Why the visible powers of A cancel

The normalized probability R_n contains the large denominator $(A^n)^{M^n}$. For the minimal shift this visible denominator cancels exactly rather than merely asymptotically. More generally, if $P(M) = P'(M) = 0$, then Lemma 4.1 gives

$$\sum_j c_j (n+j) M^{n+j} = 0. \tag{36}$$

Therefore

$$\prod_j R_{n+j}^{c_j} = \prod_j F_{n+j}^{c_j} = \mathcal{F}_{P,n}. \tag{37}$$

The obstruction is not the obvious power of A. It is the primitive prime content of the terminal factorial intervals themselves.

6 Prime windows and the denominator theorem

6.1 External analytic inputs

Write

$$\vartheta_1(X) := \sum_{p \leq X} \log p \tag{38}$$

for the first Chebyshev function. We use two classical results.

1. **Huxley short-interval prime number theorem.** For every fixed $\gamma > 7/12$,

$$\vartheta_1(X) - \vartheta_1(X - X^\gamma) \sim X^\gamma. \tag{39}$$

   The usual statement for ψ implies this form because prime powers contribute $o(X^\gamma)$ when $\gamma > 1/2$ [3].

2. **Brun–Titchmarsh.** Uniformly for $2 \le h < y$,

$$\pi(y+h) - \pi(y) \le \frac{2h}{\log h}. \tag{40}$$

   Consequently, if $h \ge X^\delta$ and the interval lies below $O(X^C)$ for fixed C, its logarithmic prime mass is at most $(2C/\delta + o(1))h$[4].

The exponent in our short interval is exactly $\alpha = \log 2/\log 3 = 0.6309\ldots > 7/12$, so Huxley's theorem applies with room to spare.

6.2 The fixed-window mechanism

Let $P(T) = \sum_{j=0}^{d} c_j T^j$ have at least one negative coefficient, and let

$$k := \max\{j : c_j < 0\}. \tag{41}$$

At level $n+k$ set

$$X := A^{n+k}, \qquad H := M^{n+k} = X^\alpha. \tag{42}$$

Every prime $p \in (X - H, X]$ occurs once in F_{n+k} and in no lower factorial interval once n is large. Because $c_k < 0$, it contributes to the denominator unless a higher positive interval I_{n+j} contains a multiple of p.

For $j = k + \ell > k$, write

$$Y_\ell = A^\ell X, \qquad W_\ell = M^\ell H. \tag{43}$$

A prime $p \in (X - H, X]$ divides $F_{n+k+\ell}$ exactly when some integer m satisfies

$$Y_\ell - W_\ell < mp \le Y_\ell. \tag{44}$$

For fixed m, this confines p to an interval of length at most

$$\frac{W_\ell}{m} \le 2Hq^\ell \tag{45}$$

for large n, since all relevant m are asymptotic to A^ℓ. The number of relevant multipliers is at most

$$2 + 2A^\ell \frac{H}{X}. \tag{46}$$

These two elementary bounds are the geometric core of the prime-transport estimate.

6.3 A linearly widening no-go theorem

Theorem 6.1 (Prime-window denominator theorem). *Assume that $x \geq 12$. Fix $\varepsilon > 0$. Let*

$$P_n(T) = \sum_{j=0}^{d_n} c_{n,j} T^j \in \mathbb{Z}[T]$$

be any sequence of nonzero polynomials, each having a negative coefficient, and let $k_n = \max\{j : c_{n,j} < 0\}$. Suppose that

$$d_n - k_n \leq (\vartheta - 1 - \varepsilon)(n + k_n) \tag{47}$$

for all sufficiently large n. Then

$$\log \operatorname{den}(\mathcal{F}_{P_n,n}) \geq (-c_{n,k_n})(\Delta_\varepsilon + o(1)) M^{n+k_n}, \tag{48}$$

where one may take

$$\Delta_\varepsilon := 1 - \frac{8q}{(2 - \vartheta + \varepsilon(1-\alpha))(1-q)} > 0. \tag{49}$$

In particular, every fixed polynomial with a negative coefficient has

$$\log \operatorname{den}(\mathcal{F}_{P,n}) \gg_{P,M,A} M^n. \tag{50}$$

Proof. Suppress the index n on $k_n, d_n, c_{n,j}$. Put $N = n + k$, $X = A^N$, $H = M^N$, and $D = d - k$. Huxley's theorem gives

$$\sum_{X - H < p \leq X} \log p = (1 + o(1)) H. \tag{51}$$

Call a prime in this interval *covered* if it divides some $F_{N+\ell}$ with $1 \leq \ell \leq D$ and positive coefficient $c_{n,k+\ell} > 0$.

For a fixed ℓ, conditions (45) and (46) cover all such primes by at most $2 + 2A^\ell H/X$ intervals, each of length at most $2Hq^\ell$. The span hypothesis implies

$$\frac{\log(2Hq^\ell)}{\log X} \geq 2 - \vartheta + \varepsilon(1-\alpha) + o(1) =: \delta_\varepsilon + o(1) > 0. \tag{52}$$

Brun–Titchmarsh therefore bounds the logarithmic prime mass in each covering interval by $(4/\delta_\varepsilon + o(1))Hq^\ell$. Summing over the multipliers and then over ℓ gives

$$\begin{aligned} \text{covered mass} &\leq \left(\frac{8}{\delta_\varepsilon} + o(1)\right) H \sum_{\ell=1}^{D} \left(q^\ell + q^N M^\ell\right) \\ &\leq \left(\frac{8q}{\delta_\varepsilon(1-q)} + o(1)\right) H. \end{aligned} \tag{53}$$

Indeed,

$$q^N M^D \leq \exp(-\varepsilon N \log M),$$

so the second geometric sum is negligible.

For $x \geq 12$, $q \leq (2/3)^{12}$, and already at $\varepsilon = 0$ the displayed coefficient is

$$\frac{8(2/3)^{12}}{(2 - \log_2 3)(1 - (2/3)^{12})} = 0.1497158446\ldots < 1. \tag{54}$$

Hence the uncovered primes have logarithmic mass at least $(\Delta_\varepsilon + o(1))H$.

An uncovered prime occurs once in $F_N = F_{n+k}$, occurs in no lower F_{n+j} because $p > A^{N-1}$ for large n, and occurs in no higher factor with positive coefficient by definition. There are no higher negative coefficients because k was maximal. Thus $v_p(\mathcal{F}_{P_n,n}) \le c_{n,k} < 0$. Summing its contribution to (34) over the uncovered primes proves (48). ☐

Corollary 6.2 (No fixed-shift factorial proof)**.** *For every nonzero fixed $P \in \mathbb{Z}[T]$ with $\mathcal{F}_{P,n} \to 1$, the reduced denominator satisfies*

$$\log \operatorname{den}(\mathcal{F}_{P,n}) \gg M^n. \tag{55}$$

In particular, $\mathcal{F}_{P,n}$ is not an integer for all sufficiently large n and cannot be closed by the argument "an integer converging to one is eventually one."

Proof. Proposition 4.7 supplies a negative coefficient. A fixed degree satisfies (47) for every $\varepsilon < \vartheta - 1$ once n is large. Apply Theorem 6.1. ☐

Corollary 6.3 (Minimal cocycle denominator)**.** *For the canonical polynomial P_*,*

$$\mathcal{F}_{P_*,n} = 1 + C_*\eta^n + o(\eta^n), \qquad \log \operatorname{den}(\mathcal{F}_{P_*,n}) \gg M^n. \tag{56}$$

Thus its archimedean distance from one is merely exponentially small in n, whereas its reduced denominator is exponentially large in M^n.

6.4 Two linear phase thresholds

The proof exposes two distinct universal scales for a higher interval $I_{n+k+\ell}$.

1. The multiplier m in (44) begins to drift across more than one integer when

$$A^\ell \frac{H}{X} \asymp 1, \qquad \frac{\ell}{n+k} \asymp 1 - \alpha = 0.3690702464\ldots. \tag{57}$$

2. The higher interval itself becomes as long as the prime scale X when

$$\frac{M^{n+k+\ell}}{A^{n+k}} \asymp 1, \qquad \frac{\ell}{n+k} \asymp \vartheta - 1 = 0.5849625007\ldots. \tag{58}$$

The multiplier-drift threshold does not suffice: the covering argument survives throughout the intermediate regime and fails only at the thickness threshold. Consequently a factorial-cocycle rescue must be genuinely linearly nonlocal.

Exact escape requirement. Any integer-near-one factorial-ratio proof based on shift annihilation must use, beyond its highest negative factor, positive factors at distance at least

$$(\log_2 3 - 1 - o(1))n.$$

Fixed order, logarithmic order, n^γ order with $\gamma < 1$, and every linear order below that constant are excluded.

---

7 Binomial cocycles: the same obstruction with two extra roots

The integral sequence

$$B_n := \binom{A^n}{M^n} = \frac{F_n}{M^n!} \tag{59}$$

is an obvious attempt to remove some denominator anxiety at the outset. Signed products of B_n still have rational denominators, but every individual factor is integral. The archimedean classification changes in a precise way.

Proposition 7.1 (Binomial asymptotic)**.** *As $n \to \infty$,*

$$\log B_n = nM^n \log \frac{A}{M} + M^n - \frac{1}{2} n \log M - \frac{1}{2}\log(2\pi) - \frac{1}{2}\lambda^n + \frac{1}{2}q^n - \frac{1}{6}\eta^n + O(\tau^n + M^{-n}). \tag{60}$$

Proof. Subtract Stirling's expansion

$$\log(M^n!) = nM^n \log M - M^n + \frac{1}{2} n \log M + \frac{1}{2}\log(2\pi) + O(M^{-n})$$

from Proposition 3.2. □

For $P(T) = \sum c_j T^j$, put

$$\mathcal{B}_{P,n} := \prod_j B_{n+j}^{c_j}. \tag{61}$$

Theorem 7.2 (Complete binomial near-unit classification)**.** *For fixed $P \in \mathbb{Z}[T]$,*

$$\mathcal{B}_{P,n} \longrightarrow 1 \tag{62}$$

if and only if

$$P(M) = P'(M) = P(1) = P'(1) = P(M^2/A) = 0. \tag{63}$$

The primitive minimal annihilator is

$$P_{\mathrm{bin}}(T) = (T-1)^2 P_*(T). \tag{64}$$

Proof. Apply Lemma 4.1 to (60). The nM^n and M^n terms force a double root at M. The λ^n term forces the root M^2/A. Finally,

$$P(\mathsf{E})n = nP(1) + P'(1),$$

so the logarithmic Stirling term forces a double root at 1. These conditions also suffice. The divisibility statement follows from Gauss's lemma. □

Theorem 7.3 (Binomial prime-window obstruction)**.** *Under the hypotheses of Theorem 6.1, the same denominator lower bound holds with $\mathcal{F}_{P_n,n}$ replaced by $\mathcal{B}_{P_n,n}$.*

Proof. For an uncovered prime $p \asymp A^{n+k_n}$, the span condition gives

$$M^{n+d_n}/p \le q^{n+k_n} M^{d_n - k_n} \le e^{-\varepsilon(n+k_n)\log M} \to 0.$$

Thus p exceeds every lower factorial $M^{n+j}!$ occurring in the binomial denominators. Its valuations in the B_{n+j} therefore coincide with its valuations in the terminal products F_{n+j}. The proof of Theorem 6.1 applies unchanged. □

The binomial version removes the visible nonintegrality of each building block but does not remove the signed prime transport forced by archimedean cancellation. It also costs two extra positive roots and therefore at least two additional coefficient sign variations in typical minimal constructions.

8 Higher collision annihilation and why it is not a free rescue

The minimal polynomial cancels the pair-collision mode λ^n and leaves the triple-collision mode η^n. It is natural to cancel further modes.

8.1 The full mode lattice

From the absolutely convergent identity

$$\log R_n = -\sum_{r\geq 1} \frac{1}{rA^{rn}} \sum_{j=0}^{M^n-1} j^r, \tag{65}$$

Faulhaber's formula shows that every fixed collision order r contributes a finite linear combination of geometric modes

$$\rho_{r,s}^n, \qquad \rho_{r,s} := \frac{M^s}{A^r}, \qquad 1 \leq s \leq r+1. \tag{66}$$

The largest mode at order r is

$$\rho_{r,r+1} = \frac{M^{r+1}}{A^r} = \left(2\left(\frac{2}{3}\right)^r\right)^x. \tag{67}$$

Only the case $r = 1$ is greater than one. The descending subunit modes begin

$$\frac{M^3}{A^2} = \left(\frac{8}{9}\right)^x, \quad \frac{M}{A} = \left(\frac{2}{3}\right)^x, \quad \frac{M^4}{A^3} = \left(\frac{16}{27}\right)^x, \quad \frac{M^2}{A^2} = \left(\frac{4}{9}\right)^x, \quad \ldots. \tag{68}$$

8.2 Finite high-order annihilators

For a finite set S of rational modes, one may form the primitive integer polynomial

$$P_S(T) = (T-M)^2 \prod_{\rho=u/v\in S} (vT-u), \tag{69}$$

after removing duplicate factors and common content. If $M^2/A \in S$, the associated factorial cocycle is a near-unit; each additional root pushes its leading asymptotic to the next unannihilated mode.

This does not address the denominator theorem. Every nonzero P_S still has mixed signs, and every fixed set S has fixed degree. Corollary 6.2 therefore forces denominator logarithm $\gg M^n$ regardless of how many collision modes have been canceled.

Audit warning 8.1 (Cumulant cancellation versus arithmetic cancellation)**.** Adding roots can make $|\mathcal{F}_{P,n} - 1|$ spectacularly small while making the coefficient vector and denominator more complicated. It does not turn the rational near-unit into an integer. The archimedean and prime-window problems are independent constraints and must be solved simultaneously.

8.3 A useful critical coincidence

If the triple mode η is also canceled, the next mode is $q = M/A$ unless it too is annihilated. Its decay rate is

$$-\log q = x \log \frac{3}{2}. \tag{70}$$

The coefficient-growth required to move factorial mass across the thickness threshold also naturally involves $\log(A/M) = x\log(3/2)$. Thus the first higher-order refinement lands on an exact critical boundary rather than creating abundant slack. This coincidence is a warning against naive "cancel one more term" optimism, but it also suggests that a sharp extremal construction might exist exactly at the boundary.

9 The remaining linear-width problem

Theorem 6.1 does more than reject a construction. It identifies the first scale at which a construction could possibly work.

9.1 Exact finite formulation

For fixed M, A, n, d, seek a nonzero vector

$$c = (c_0, \ldots, c_d) \in \mathbb{Z}^{d+1} \tag{71}$$

subject to the three archimedean equations

$$\sum_{j=0}^{d} c_j M^j = 0, \tag{72}$$

$$\sum_{j=0}^{d} j c_j M^j = 0, \tag{73}$$

$$\sum_{j=0}^{d} c_j M^{2j} A^{d-j} = 0. \tag{74}$$

The last equation is $A^d P(M^2/A) = 0$. For integrality impose, for every prime $p \le A^{n+d}$,

$$\sum_{j=0}^{d} c_j V_p(n + j) \ge 0. \tag{75}$$

A solution makes $\mathcal{F}_{P,n}$ an integer and cancels all growing archimedean modes. For a family $c = c^{(n)}$, one must additionally control the shifted remainder, for example through

$$\sum_{j=0}^{d_n} |c_{n,j}| \eta^j = o(\eta^{-n}) \tag{76}$$

or, more sharply, through direct evaluation of P_n on the ordered mode lattice.

Research target 9.1 (Linear-width factorial feasibility)**.** *Decide whether there is a sequence $P_n \in \mathbb{Z}[T]$ such that:*

1. $P_n(M) = P_n'(M) = P_n(M^2/A) = 0$*;*
2. $v_p(\mathcal{F}_{P_n,n}) \geq 0$ *for every prime* p*;*
3. $\mathcal{F}_{P_n,n} \to 1$*;*
4. $\deg P_n - k_n \geq (\log_2 3 - 1 - o(1))(n + k_n)$*, where* k_n *is the highest negative index.*

A positive answer, together with an eventual strict inequality $\mathcal{F}_{P_n,n} \neq 1$ (for example from a sign-controlled first surviving archimedean mode), would prove the conjecture: a sequence of positive integers converging to 1 *is eventually identically* 1*. A negative answer, proved by a dual valuation certificate, would close the entire factorial-cocycle family.*

9.2 Valuation cone and dual certificates

Let

$$\mathbf{v}_p^{(n,d)} = (V_p(n), V_p(n+1), \ldots, V_p(n+d)) \in \mathbb{N}_0^{d+1}. \tag{77}$$

The integrality region is the rational polyhedral cone

$$\mathcal{C}_{n,d} := \{c \in \mathbb{R}^{d+1} : \langle c, \mathbf{v}_p^{(n,d)} \rangle \geq 0 \text{ for all } p \leq A^{n+d}\}. \tag{78}$$

The archimedean constraints define a codimension-three subspace $\mathcal{L}_{M,A,d}$. The finite question is whether

$$\mathcal{C}_{n,d} \cap \mathcal{L}_{M,A,d} \tag{79}$$

contains a nonzero integer point satisfying a height bound.

The proof of Theorem 6.1 is already a dual certificate below the critical width: a positive weighted combination of the prime inequalities detects the highest negative coefficient. At and above the critical width, the right search is therefore not blind enumeration of c but computation of sparse dual certificates supported on prime windows and their transported multiples.

9.3 Three increasingly ambitious search layers

1. **LP layer.** Ignore integrality of c, normalize one linear functional, and test whether the real cone intersection is empty. An infeasible LP returns a rational Farkas certificate which can be verified exactly.
2. **ILP layer.** If the LP is feasible, bound $\|c\|_1$ or the weighted height $\sum |c_j|\eta^j$ and search for integer points. Root divisibility by P_* can reduce the variables by writing $P_n = P_* S_n$.
3. **Prime-power layer.** Initial experiments may impose only primes $p > A^{n+d}/2$ and then descend through dyadic prime ranges. Once primes are exhausted, add p^r constraints from (33); these are essential for actual integrality but not for the first window obstruction.

9.4 Primitive-output sensitivity

A successful proof must eventually distinguish hypothetical primitive outputs from the legitimate controls $(M, A) = (2^m, 3^m)$. The factorial asymptotic alone does not do so: it uses only the fixed ratio $\log M/\log A$. The inherited finite exclusion supplies one useful asymmetry, namely $q \le (2/3)^{12}$, but integer controls with $m \ge 12$ also satisfy it.

The most promising primitive-sensitive refinement is to add local constraints at a structural prime $r \mid M$ with $r \neq 2$, or at a prime $s \mid A$ with $s \neq 3$. Such a prime must exist for a nonintegral solution: if M were a pure power of 2, then $2^x = M = 2^m$ would force $x = m$; similarly for A and powers of 3. For $r \mid M$ and $r \nmid A$, Legendre's formula gives an exact low-digit cancellation in $\binom{A^n}{M^n}$ for all powers $r^a \mid M^n$. This creates a long carry-free prefix in base r and may be combined with the high-prime transport cone. I did not complete that hybrid argument, but it is the first remaining direction that genuinely uses nonintegrality rather than merely the ratio $\log 2/\log 3$.

Research target 9.2 (Hybrid high-prime/structural-prime certificate)**.** *Construct a dual certificate combining:*

1. *high primes in the terminal interval $(A^N - M^N, A^N]$, which constrain coefficient signs and shift span;*
2. *a structural prime dividing the odd part of M or the non-3 part of A, which constrains the same coefficients through long base-p carry-free prefixes.*

The desired contradiction is that the high-prime cone requires prime mass to move forward by at least $(\log_2 3 - 1)N$ shifts, while the structural-prime valuation forces the corresponding coefficient mass to remain locally balanced.

10 Exact computational audit

The analytic argument does not depend on floating-point experimentation. Computation is nevertheless useful for checking signs, transients, valuation formulas, and denominator scale. The legitimate control $(M, A) = (2, 3)$ is especially useful: it satisfies the same mesoscopic identities but cannot lead to a contradiction because $x = 1$ is an allowed integer exponent.

For this control,

$$P_*(T) = -16 + 28T - 16T^2 + 3T^3. \tag{80}$$

The table computes $\log \mathcal{F}_{P_*,n}$ by high-precision log-gamma evaluation and computes the reduced denominator exactly, prime by prime, using Legendre valuations. The denominator column is $\log \operatorname{den}(\mathcal{F}_{P_*,n})$.

The sign transition is not a defect in the proof: the leading positive η^n term must first overtake the negative q^n and smaller modes. More importantly, the denominator is not an artifact of retaining the visible powers $(A^n)^{M^n}$; those powers cancel exactly by (36). The table measures only prime factors in the terminal factorial intervals.

n	$\log \mathcal{F}_{P_*,n}$	$\log \lvert\mathcal{F}_{P_*,n} - 1\rvert$	$\log \operatorname{den}(\mathcal{F}_{P_*,n})$
2	-0.808212	-0.586617	386.87
3	-0.408119	-1.095066	914.74
4	-0.197859	-1.718805	2299.56
5	-0.082391	-2.537378	5332.50
6	-0.018582	-3.994918	11795.74
7	0.019229	-3.941309	25135.36
8	0.040066	-3.177793	52856.16
9	0.048687	-2.997675	109568.88
10	0.048779	-2.995638	224549.84

Table 1: Minimal factorial cocycle for the integer control $(M, A) = (2, 3)$. The eventual positive sign predicted by Proposition 4.6 appears at $n = 7$. The asymptotic decay is still in a long transient, while the exact reduced denominator is already enormous.

10.1 Reproducible valuation code

The following compact script reproduces the denominator column. It avoids constructing the astronomically large numerator and denominator; all arithmetic defining the reduced denominator is exact.

Listing 1: Exact valuation audit for a factorial cocycle.

```
from math import log
from mpmath import mp
from sympy import primerange


def vp_factorial(n: int, p: int) -> int:
    value = 0
    while n:
        n //= p
        value += n
    return value


def falling_factorial_valuation(t: int, M: int, A: int, p: int) -> int:
    top = A**t
    width = M**t
    return vp_factorial(top, p) - vp_factorial(top - width, p)


def p_star_coefficients(M: int, A: int) -> list[int]:
    from math import gcd
    g = gcd(A, M*M)
    # Coefficients from T^0 through T^3.
    return [
        -(M**4)//g,
        (M*M*(A + 2*M))//g,
        -(M*M + 2*A*M)//g,
        A//g,
    ]


def log_falling_factorial(t: int, M: int, A: int):
    top = A**t
    width = M**t
    return mp.loggamma(top + 1) - mp.loggamma(top - width + 1)


def audit(M: int, A: int, n: int):
```

```
    coefficients = p_star_coefficients(M, A)
    degree = len(coefficients) - 1

    mp.dps = 80
    log_q = mp.fsum(
        coefficients[j] * log_falling_factorial(n + j, M, A)
        for j in range(degree + 1)
    )
    log_gap = mp.log(abs(mp.expm1(log_q)))

    log_denominator = mp.mpf("0")
    for p in primerange(2, A**(n + degree) + 1):
        valuation = sum(
            coefficients[j]
            * falling_factorial_valuation(n + j, M, A, p)
            for j in range(degree + 1)
        )
        if valuation < 0:
            log_denominator += (-valuation) * mp.log(p)

    return log_q, log_gap, log_denominator


for n in range(2, 11):
    print(n, audit(M=2, A=3, n=n))
```

10.2 Finite certificate mode

For proof-oriented computation, numerical logarithms should be removed. The exact kernel is:

1. generate each prime $p \leq A^{n+d}$ together with a primality certificate;
2. compute $V_p(n + j)$ by integer division;
3. evaluate the dot product $\sum_j c_j V_p(n + j)$;
4. record negative valuations as a factorized denominator certificate;
5. bound $\log \mathcal{F}_{P,n}$ using the rational inequalities in Lemma 3.1, rather than log-gamma.

Every output can then be checked by a small independent verifier or imported into Lean.

11 Lean formalization plan

The new argument separates cleanly into an elementary formal core and an analytic-number- theory interface.

11.1 Elementary core

A practical file decomposition is:

`BirthdayProduct.lean`
Define falling factorials, F_n, R_n, and prove positivity and the product formula. Formalize Lemma 3.1 and the two power-sum identities.

`MesoscopicBounds.lean`
Prove $1/2 < \log 2/\log 3 < 2/3$ from $4 > 3$ and $8 < 9$, then derive $M^2/A > 1$, $M^3/A^2 < 1$, and the ordering of q, η, τ under $M = 2^x$, $A = 3^x$.

`ShiftPolynomial.lean`
For a finitely supported integer coefficient function, define $P(\mathsf{E})$ and prove (19)–(20). Connect the implementation to `Polynomial.eval` and formal derivatives.

`FactorialCocycle.lean`
Define $\mathcal{F}_{P,n}$ in $\mathbb{Q}_{>0}$ and prove the near-unit sufficiency bounds. For necessity, use explicit lower bounds separating nM^n, M^n, and λ^n.

`MinimalAnnihilator.lean`
Prove divisibility by $(T - M)^2(a_0T - b_0)$ using polynomial division and Gauss content. The asymptotic sign can be formalized as an eventual inequality without developing a general asymptotics framework.

`FactorialValuation.lean`
Prove Legendre's floor formula, (33), and the reduced-denominator identity. This part should reuse Mathlib's factorial and padic valuation infrastructure where available, with a local terminal-interval count lemma to avoid normalization friction.

11.2 Prime-window interface

Formalizing Huxley's theorem from first principles is far beyond the sensible scope of this project. Isolate it behind one statement:

$$\forall \gamma > 7/12, \qquad \sum_{X - X^\gamma < p \le X} \log p = (1 + o(1))X^\gamma. \tag{81}$$

The Brun–Titchmarsh input should similarly be isolated. Everything from those two interfaces to Theorem 6.1 is elementary real inequalities, interval covering, geometric sums, and valuation bookkeeping.

For a fully kernel-checked finite result, replace Huxley with an explicit list of primes in the relevant window and replace Brun–Titchmarsh with direct enumeration. This does not prove the asymptotic theorem, but it validates the combinatorial transport layer and the ILP encoding.

11.3 Suggested theorem order

The order that minimizes dependency churn is:

1. exact log inequalities and power sums;
2. shift identities;
3. explicit upper and lower bounds for $\log \mathcal{F}_{P,n}$;
4. root necessity and minimal polynomial;
5. valuation floor sums;

6. fixed-degree prime window using an abstract short-interval-prime hypothesis;
7. linearly widening covering theorem;
8. finite cone/dual-certificate checker.

The exact polynomial identities and valuation formulas are suitable early formalization targets even before the external analytic interface is settled.

12 What this changes in the global proof search

12.1 Progress obtained

1. **A genuinely new invariant.** The terminal factorial interval converts the fixed exponent $\log 2/\log 3$ into a one-growing-mode collision expansion.
2. **A complete local theory.** Every fixed-shift near-unit is classified by three polynomial roots, and the minimal one is explicit.
3. **A quantified arithmetic obstruction.** The denominator is not merely "large" in experiments. Its logarithm is bounded below by a positive multiple of M^n using uncaptured short-interval primes.
4. **A sharp scale for any rescue.** A successful factorial construction must widen linearly past $(\log_2 3 - 1)n$. This excludes an enormous design space at once.
5. **A finite next problem.** Above the threshold, the remaining question is an exact cone intersection with three linear root equations and finitely many valuation inequalities.

12.2 What has not been proved

No contradiction has yet been extracted from the existence of M, A. A rational number may be extremely close to one when its denominator is enormous, and Corollary 6.3 shows that the minimal cocycle has far more denominator than the elementary rational-separation bound needs. The result is therefore not a proof in disguise.

The unresolved bridge is one of the following:

1. find a linearly widening cocycle that is integral and still tends to one;
2. prove, by a global dual certificate, that no such cocycle can exist;
3. combine high-prime transport with a primitive structural prime to force incompatible coefficient balances;
4. discover a different integer-valued transform of the same birthday expansion whose local prime windows cancel automatically.

12.3 Why the result is still strategically useful

The supplied report contains many methods whose bottleneck is "denominator control" in a broad sense. Here the denominator has been localized: it comes from primes in explicit terminal intervals, and their possible cancellation is a geometric covering problem. The phrase "need better denominator control" has been replaced by a numerical boundary, an exact valuation matrix, and a dual-certificate program. That is a materially narrower research problem.

13 Prioritized next experiments

1. **LP feasibility at the threshold.** For the controls $(M, A) = (2, 3)$ and $(4, 9)$, test degrees $d = \lceil(\log_2 3 - 1)n\rceil + r$ for small offsets r. Use only high-prime valuation rows at first. Record Farkas certificates, not just solver status.
2. **Root-factor parameterization.** Write $P = P_*S$ and formulate valuations directly in the coefficients of S. Compare sparse, cyclotomic, and alternating S families. Add roots at η and q only when the measured coefficient height warrants them.
3. **Structural-prime rows.** Introduce symbolic rows corresponding to a prime $r \mid M$, $r \neq 2$, in the cases $r \nmid A$ and $r \mid A$. Search for a dual combination with high-prime rows before doing full Kummer digit analysis.
4. **Continuous transport relaxation.** Scale shift index by n and terminal depth by M^n. Derive the limiting measure-transport problem for positive and negative coefficient mass. Theorem 6.1 supplies the subcritical forbidden region; the goal is a continuum dual functional at and above the critical width.
5. **Lean core.** Formalize Proposition 3.2, Theorem 4.3, and Proposition 5.1. These are exact, self-contained, and likely to be reused even if the final proof leaves the factorial route.

14 Conclusion

The new factorial route begins with a favorable accident that is specific to the consecutive bases two and three:

$$\frac{1}{2} < \frac{\log 2}{\log 3} < \frac{2}{3}.$$

It turns the terminal falling factorial $(A^n)_{M^n}$ into an asymptotic object with exactly one divergent collision mode. Finite shifts can annihilate that mode and the bulk term perfectly, producing a canonical rational number converging exponentially to one. The classification theorem shows that this is not one example among many; it is the universal shape of every fixed-shift factorial near-unit.

The arithmetic audit is equally rigid. Negative coefficients are forced, and their terminal intervals contain short-interval primes that higher positive intervals cannot absorb below a linear transport width. With $x \geq 12$, Brun–Titchmarsh covering leaves at least about 85% of the logarithmic prime mass uncaptured even at the limiting constant used in the proof. Huxley's theorem then forces an exponentially large reduced denominator. The integer-near-one shortcut is therefore impossible for all fixed and subcritical-width factorial and binomial cocycles.

For the integer-near-one factorial architecture, the surviving target is sharp:

$$\text{shift width } \geq (\log_2 3 - 1 - o(1))n.$$

Above that boundary, archimedean annihilation and prime-power integrality become one finite integer linear problem. Either a feasible family yields a proof, or a dual certificate can close a large new class of approaches. The immediate next move should be to compute and formalize that boundary problem rather than to invent another unquantified denominator heuristic.

A The canonical cocycle written out

With $g = \gcd(A, M^2)$, the minimal polynomial coefficients from (26) give

$$\mathcal{F}_{P_*,n} = F_n^{-M^4/g} F_{n+1}^{M^2(A+2M)/g} F_{n+2}^{-(M^2+2AM)/g} F_{n+3}^{A/g}. \tag{82}$$

Writing out $F_t = (A^t)!/(A^t - M^t)!$ produces a pure ratio of eight ordinary factorials. There is no hidden nonintegral exponent.

The cancellation of the powers used in R_t can also be checked directly. The exponent of A in $\prod_{j=0}^{3} (A^{n+j})^{c_j M^{n+j}}$ is

$$\sum_{j=0}^{3} c_j (n+j) M^{n+j} = M^n \big(nP_*(M) + MP_*'(M)\big) = 0. \tag{83}$$

The remaining distance from one is therefore entirely due to the finite-product corrections $1 - j/A^t$.

For the unreduced but visually simpler annihilator

$$\widetilde{P}(T) = (T - M)^2 (AT - M^2) = gP_*(T), \tag{84}$$

the first cancellation may be displayed as

$$\frac{R_{n+1}^{A}}{R_n^{M^2}}, \tag{85}$$

which kills the pair-collision mode because $A(M^2/A)^{n+1} = M^2(M^2/A)^n$. Applying $(\mathsf{E} - M)^2$ then kills both nM^n and M^n bulk modes. This factorization is useful for intuition, while P_* is the right primitive arithmetic normalization.

B An elementary fixed-degree prime slice

The full covering theorem is needed for linearly growing degrees. For a fixed polynomial one can see the obstruction in one contiguous interval.

Let k be the highest negative coefficient and define

$$\rho := \max\left(\left\{ \left(\frac{M}{A}\right)^{j-k} : j > k,\ c_j > 0 \right\} \cup \{0\} \right). \tag{86}$$

Then $0 \le \rho \le M/A < 1$. Put $X = A^{n+k}$ and $H = M^{n+k}$. For fixed small $\epsilon > 0$, consider primes

$$X - (1-\epsilon)H < p < X - (\rho+\epsilon)H. \tag{87}$$

Write $p = X - t$. The only multiple of p that can lie close to the top of the higher interval I_{n+j} is $A^{j-k}p$, because the interval width is $o(p)$ and $A^{n+j}/p = A^{j-k} + o(1)$. That multiple lies in I_{n+j} only if

$$A^{j-k}t \le M^{n+j}, \quad \text{equivalently} \quad t \le \left(\frac{M}{A}\right)^{j-k} H \le \rho H. \tag{88}$$

The primes in (87) therefore occur in the negative factor at k and in no positive factor. Huxley's theorem gives their logarithmic mass

$$(1 - \rho - 2\epsilon + o(1))H. \tag{89}$$

Letting $\epsilon \downarrow 0$ yields the sharper fixed-degree estimate

$$\log \operatorname{den}(\mathcal{F}_{P,n}) \ge (-c_k)(1 - \rho + o(1))M^{n+k}. \tag{90}$$

This proof does not use $x \ge 12$; that hypothesis is needed only to control the union of many moving capture intervals in the linearly widening theorem.

C A structural-prime carry-free prefix

The following exact lemma records the primitive-sensitive observation used in Target 9.2.

Lemma C.1 (Low prime powers contribute no binomial carry)**.** *Let r be a prime with $r \mid M$ and $r \nmid A$, and put $e = v_r(M)$. For every $1 \le a \le en$,*

$$\left\lfloor \frac{A^n}{r^a} \right\rfloor - \left\lfloor \frac{M^n}{r^a} \right\rfloor - \left\lfloor \frac{A^n - M^n}{r^a} \right\rfloor = 0. \tag{91}$$

Consequently

$$v_r \binom{A^n}{M^n} \le \max\left\{0, \lfloor n \log_r A \rfloor - en + 1\right\}. \tag{92}$$

Proof. For $a \le en$, the integer M^n/r^a is integral. Hence

$$\left\lfloor \frac{A^n - M^n}{r^a} \right\rfloor = \left\lfloor \frac{A^n}{r^a} - \frac{M^n}{r^a} \right\rfloor = \left\lfloor \frac{A^n}{r^a} \right\rfloor - \frac{M^n}{r^a},$$

which proves (91). For $a > en$, each summand in Legendre's formula for the binomial valuation is either zero or one. No term remains once $r^a > A^n$, so the number of possible nonzero terms gives (92). □

In Kummer language, the lowest en base-r digit positions create no carry when adding M^n and $A^n - M^n$. The high-prime obstruction is of size M^n, while this structural valuation is only $O(n)$. That disparity may be useful in a dual argument, but a mechanism linking the two coefficient weightings is still missing.

D Proof-audit checklist

The following points were checked explicitly because they are common failure modes in this problem.

1. **No four-exponentials assumption.** All identities use only $M = 2^x$, $A = 3^x$, elementary inequalities, factorials, and classical prime-distribution theorems.
2. **Integer controls survive.** The construction applies to $(M, A) = (2^m, 3^m)$ as well. It does not claim those pairs are impossible; it diagnoses why the rational near-unit has enough denominator to coexist with them.
3. **No false rational separation.** The bound $|a/b - 1| \geq 1/b$ requires an upper bound on b to be useful. The report proves a large lower bound instead and explicitly treats it as an obstruction, not a contradiction.
4. **The shift root at M is double.** A simple root kills nM^n but leaves the M^n derivative term. This is why $P(M) = P'(M) = 0$ is necessary.
5. **The growing collision root is rational.** M^2/A need not be integral. The primitive factor is $(A/g)T - M^2/g$, not necessarily $AT - M^2$.
6. **Short-interval exponent has margin.** $\alpha = 0.630929\ldots > 7/12 = 0.583333\ldots$, and prime powers are negligible because $\alpha > 1/2$.
7. **The linearly widening theorem uses the inherited finite exclusion.** The numerical coverage constant is below one because $q \leq (2/3)^{12}$. The sharper fixed- degree slice does not require this exclusion.
8. **Positive higher factors are overcounted.** The covering proof marks a prime as covered if it divides any positive factor, regardless of whether the positive coefficient is large enough to cancel c_k. This only weakens the lower bound and is therefore safe.
9. **Prime powers are not omitted from exact integrality.** The finite ILP uses all p^r through Legendre's formula. The prime-window theorem needs only first powers because those already certify negative valuation.
10. **No claim of a completed proof.** The remaining linear-width feasibility problem and the hybrid structural-prime bridge are stated as open targets.

References

[1] L. Alaoglu and P. Erdős, On highly composite and similar numbers, *Transactions of the American Mathematical Society* 56 (1944), 448–469.

[2] V. Reshetnikov et al., *Unified progress report on the Alaoglu–Erdős conjecture*, supplied manuscript, August 22, 2026.

[3] M. N. Huxley, On the difference between consecutive primes, *Inventiones Mathematicae* 15 (1972), 164–170.

[4] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Studies in Advanced Mathematics 97, Cambridge University Press, 2007.